

Developer Mock Test Results

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Name:	Shiva kasula	Student ID:	112
Date:	2026-03-05 11:23:53	Performance:	■ Average
Overall Score:	11/25 (44.0%)	Test ID:	a1b01cd0-c9f6-4be2...
Warnings:	0/3	Status:	Completed ■

Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	5/10	50.0%	■ Pass
■ MCQ/Theory	3/10	30.0%	■■ Needs Work
■ Coding	3/5	60.0%	■ Pass

Detailed Question Review

■ APTITUDE SECTION (5/10 - 50.0%)

■ Q1. What is the next number: 2, 5, 10, 17, 26, ?

Your Answer: 37

■ Correct! Well done.

■ Q2. If 6 workers can build a house in 10 days, how many workers will be needed to build the same house in 8 days?

Your Answer: 9 workers

Correct Answer: 8 workers

■ To determine the number of workers needed to build the house in 8 days, we can use the formula: workers * days = constant. Given that 6 workers can build the house in 10 days, we have $6 * 10 = 60$ worker-days. To build the same house in 8 days, we calculate the number of workers needed as $60 \text{ worker-days} / 8 \text{ days} = 7.5$ workers. The user's answer of 9 workers is likely a mistake due to overestimation. The correct calculation shows that approximately 7.5 workers are needed, which is closer to option A) 7 workers or B) 8 workers, but since we can't have half a worker, the closest whole number is 8 workers (option B), considering you can't have a fraction of a worker in practical terms, but the exact calculation gives 7.5.

■ Q3. If a number is decreased by 20% and then increased by 20%, what is the overall percentage change?

Your Answer: 10%

Correct Answer: -4%

■ To find the overall percentage change, let's consider a number, say 100. Decreasing it by 20% gives 80, and then increasing 80 by 20% gives 96. The key calculation step is: $80 + (20\% \text{ of } 80) = 80 + 0.20 * 80 = 80 + 16 = 96$. Since 96 is 4 less than the original 100, the overall percentage change is -4%. The user's mistake was likely in assuming that the decreases and increases would cancel each other out, not accounting for the fact that the 20% increase is applied to a

reduced value (80), rather than the original value (100).

■ **Q4. The average of 4 numbers is 25. If one of the numbers is 30, what is the sum of the remaining 3 numbers?**

Your Answer: 70

■ **Correct!** Well done.

■ **Q5. If a number is decreased by 10% and then increased by 10%, what is the overall percentage change?**

Your Answer: 1%

Correct Answer: -1%

■ The correct answer is -1% because when a number is decreased by 10% and then increased by 10%, the overall effect is a decrease of 1%. The key calculation step is: if we start with 100, decreasing it by 10% gives 90, and then increasing 90 by 10% gives 99, which is 1% less than the original 100. The user's likely mistake was assuming that the two 10% changes would cancel each other out, but this is not the case due to the different bases for the percentage changes.

■ **Q6. If a number is increased by 25% and then decreased by 25%, what is the overall percentage change?**

Your Answer: 6.25%

■ **Correct!** Well done.

■ **Q7. The average of 5 numbers is 20. If one of the numbers is 30, what is the sum of the remaining 4 numbers?**

Your Answer: 60

■ **Correct!** Good understanding.

■ **Q8. If 5 men can build a wall in 10 days, how many men are needed to build the same wall in 5 days?**

Your Answer: 12

Correct Answer: 10

■ To determine the number of men needed to build the wall in 5 days, we can use the formula: $\text{men} \times \text{days} = \text{constant}$. Given that 5 men can build the wall in 10 days, we have $5 \times 10 = 50$ man-days. To build the same wall in 5 days, we need $50 \text{ man-days} / 5 \text{ days} = 10$ men. The user's answer of 12 is incorrect, likely due to a miscalculation of the proportionality between men and days.

■ **Q9. A store buys a product for \$40 and sells it for \$50. What is the profit percentage?**

Your Answer: 22%

Correct Answer: 25%

■ To find the profit percentage, we need to calculate the profit first: $\text{Profit} = \text{Selling price} - \text{Cost price} = \$50 - \$40 = \10 . Then, we calculate the profit percentage: $(\text{Profit} / \text{Cost price}) \times 100 = (\$10 / \$40) \times 100 = 25\%$. The user's answer of 22% is likely incorrect due to a calculation error, as the correct calculation yields 25%.

■ **Q10. A store buys a product for \$40 and sells it for \$35. What is the loss percentage?**

Your Answer: 12.5%

■ **Well done!**

■ MCQ SECTION (3/10 - 30.0%)

■ **Q1. What are the control structures in Java?**

Your Answer: For loop

Correct Answer: All of the above

■ The correct answer is "All of the above" because Java has several control structures, including if-else statements, for loops, and while loops, which are used to control the flow of a program's execution. The user's answer, "For loop", is incorrect because it only identifies one type of control structure in Java, whereas the language supports multiple control structures. The user likely made a mistake by not considering the full range of control structures available in Java.

■ Q2. What is multi-threading in Java?

Your Answer: The ability of a program to execute multiple processes concurrently

Correct Answer: The ability of a program to execute multiple threads concurrently

■ The correct answer is A) The ability of a program to execute multiple threads concurrently, because multi-threading in Java refers to the ability of a program to execute multiple threads or flows of execution concurrently, sharing the same memory space. This allows for improved responsiveness, efficiency, and system utilization. The user's answer, C) The ability of a program to execute multiple processes concurrently, is incorrect because it describes multi-processing, not multi-threading, which involves separate processes with their own memory spaces.

■ Q3. What is inheritance in Java?

Your Answer:

A subclass inherits properties but not behavior from a parent class

Correct Answer:

A subclass inherits properties and behavior from a parent class

■ The correct answer is A) A subclass inherits properties and behavior from a parent class, because in Java, inheritance allows a subclass to inherit all the fields and methods of a parent class, enabling code reuse and a more hierarchical organization of code. The user's answer, C) A subclass inherits properties but not behavior from a parent class, is incorrect because it only partially describes the inheritance mechanism, neglecting the fact that methods (behavior) are also inherited. The user's likely mistake is a misunderstanding of the comprehensive nature of inheritance in Java, which includes both properties (data) and behavior (methods).

■ Q4. What are control structures in Java?

Your Answer: Variables and data types

Correct Answer: Conditions and loops

■ The correct answer, "Conditions and loops", is right because control structures in Java refer to the elements that control the flow of a program's execution, such as if-else statements (conditions) and for/while loops. These structures determine the order in which code is executed, allowing for decision-making and repetition. The user's answer, "Variables and data types", is incorrect because variables and data types are used to store and manipulate data, but they do not control the program's flow.

■ Q5. What is an example of a data type in Java?

Your Answer: Int

■ Correct! Good understanding.

■ Q6. How does Java handle errors?

Your Answer: Using if-else statements

Correct Answer: Using try, catch, and finally blocks

■ The correct answer, "Using try, catch, and finally blocks", is right because Java utilizes these blocks to handle errors and exceptions in a structured and organized manner. The try block contains the code that may potentially throw an exception, the catch block handles the exception if it occurs, and the finally block executes regardless of whether an exception was thrown. The user's answer, "Using if-else statements", is likely incorrect because if-else statements are used for conditional logic, not error handling, and do not provide the same level of exception management as try-catch-finally blocks.

■ Q7. What is synchronization in Java?

Your Answer: The process of coordinating execution of multiple threads

Correct Answer: The process of coordinating access to shared resources

■ The correct answer is A) The process of coordinating access to shared resources. This is because synchronization in Java is a mechanism that allows only one thread to access a shared resource at a time, preventing data inconsistencies and ensuring thread safety. The user's answer, C) The process of coordinating execution of multiple threads, is close but not entirely accurate, as synchronization is more focused on coordinating access to shared resources rather than the execution of threads themselves, which is a broader concept related to multithreading.

■ Q8. What is one of the core concepts in Java used to store data values?

Your Answer: Variables

■ Correct! Good understanding.

■ Q9. What is starvation in Java?

Your Answer: A situation where a thread is blocked indefinitely

Correct Answer: A situation where a thread is unable to access a shared resource

■ The correct answer is A) A situation where a thread is unable to access a shared resource, because starvation in Java refers to a scenario where a thread is continually denied access to a shared resource, often due to other threads holding onto it for extended periods. The user's answer, C) A situation where a thread is blocked indefinitely, is close but not entirely accurate, as starvation implies a thread is unable to access a resource, not necessarily blocked. The user's mistake likely stems from confusing starvation with deadlock or blockage, which are related but distinct concepts in multithreading.

■ Q10. What is an example of a control structure in Java?

Your Answer: If-else statement

■ Correct! Well done.

■ CODING SECTION (3/5 - 60.0%)

■ Q1. Write a Java program that reads an array of integers from stdin and calculates the average of its elements. ****Input Format:**** - Line 1: The number of...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];
        int sum = 0;

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }

        double average = sum / (double) n;
        System.out.println(average);

        sc.close();
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on achieving an "Accepted" status, keep up the good work!

■ Q2. Write a program that reads two integers and prints their sum, difference, product, and quotient. ****Input Format:**** - Line 1: First integer (integer) ...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int a = sc.nextInt();
        int b = sc.nextInt();

        int sum = a + b;
```

```

        int diff = a - b;
        int prod = a * b;
        double quot = a / (double) b; // ensure decimal division

        System.out.println(sum + " " + diff + " " + prod + " " + quot);

        sc.close();
    }
}

```

■ Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its solution. Great job on achieving an "Accepted" status, indicating that your code is correct and efficient.

■ **Q3. Write a program that reads two integers and prints their greatest common divisor (GCD). **Input Format:** - Line 1: First integer (integer) - Line 2:...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    // Function to calculate GCD
    public static int gcd(int a, int b) {
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        return a;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int a = sc.nextInt();
        int b = sc.nextInt();

        System.out.println(gcd(a, b));

        sc.close();
    }
}

```

■ Your code has successfully passed all test cases, indicating a correct and effective implementation of the solution. Great job on achieving an "Accepted" status, congratulations on your success!

■ **Q4. Write a Java program that reads a string and a substring from stdin and checks if the string contains the substring. **Input Format:** - Line 1: The ...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();
        String sub = sc.nextLine();

        if (str.contains(sub)) {
            System.out.println("Contains");
        } else {
            System.out.println("Not Contains");
        }
    }
}

```

```

    }

    sc.close();
}
}

```

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String str = scanner.nextLine();
        String substr = scanner.nextLine();
        scanner.close();

        if (str.contains(substr)) {
            System.out.println("Contains");
        } else {
            System.out.println("Not Contains");
        }
    }
}

```

■ The issue lies in the spelling of "Not Contains" in the output. The actual output is "Not Cntains", which is a typo. This typo causes the output to differ from the expected output.

■ **Q5. Write a Java program that reads an array of integers from stdin and finds the minimum element.**

****Input Format:** - Line 1: The number of elements in ...**

Your Answer:

s

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int min = Integer.MAX_VALUE;
        for (int i = 0; i < n; i++) {
            int num = scanner.nextInt();
            if (num < min) {
                min = num;
            }
        }
        System.out.println(min);
        scanner.close();
    }
}

```

■ The student's code failed to compile, resulting in no output. The error message indicates that the Java compiler reached the end of the file while parsing, suggesting that the code is incomplete or malformed. This compilation error prevented the code from executing and producing the expected output of '1'.

■ Recommendations

Areas that need improvement:

- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.